

Andy Ye

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Education

University of Washington

Seattle, WA

B.S. in Computer Science - September 2024-2028. GPA: 3.63

Relevant Coursework: Deep Learning, Autonomous Robotics, Linear Optimization, Data Structures and Algorithms, Probability and Combinatorics, Systems Programming, Logic, Interaction Programming

Experience

Research Intern: Brown AI-REU - June 2026 – Aug 2026

ACT Lab

- Advised by Prof. Nora Ayanian and Aditya Iyer, working on drone swarm repositioning, ad-hoc mobile networks, and communication in heterogeneous robot systems.

Undergraduate Researcher - October 2025 – May 2026

Ubiquitous Computing Lab

- Built a bernoulli level set estimation algorithm with gaussian process models to optimize finding critical flicker frequency (CFF) threshold detection for BEACON, a mobile health diagnostic tool.
 - User diagnosis time and query count **reduced by over half** of previous methods.
- Built ElectroNome, a low power, limb-agnostic interaction system for AR/XR glasses.

Undergraduate Researcher - October 2025 – April 2026

Allen Institute for AI

- Worked with Matt Wallingford in the Perceptual Reasoning and Interaction Research team.
- Built Bristle, an automated pipeline to convert real estate videos into a CV dataset to supplement sim training, producing **20,000+ clips** with continuous motion trajectories for embodied AI navigation.
- Implemented endpoint object classification with DINO and via ray-casting for occlusions to extract walkable destination coordinates. Processed over 5TB with SLURM.

Research

- *ElectroNome: Limb-Agnostic Synchronous Gestures via Electric-Field Sensing*
Cynthia Dong, **Andy Ye**, Jared Hwang, Zachary Englhardt, Jerry Cao, Juyoung Lee, Shwetak Patel, Richard Li

Selected Projects

DAWG'D - March 2026 – June 2026

Python, Slurm

Generated a dataset of adversarially masked websites to cause vision-based webagent systems to return tampered results. We adversarially perturb 10% of the screenshot's area.

- We introduce L1 attacks, maximizing the difference between the correct answer and the incorrect. L2 is nuanced, causing MolmoWeb to return plausible lies. L3 changes facts, ie, July instead of June.
- We study partitioning the perturbed 10% into smaller chunks, and find that total mass is what allows for adversarial attacks, not how it is split.

HeLU-DTI - July 2023 – August 2024

Python, Pytorch, HGNN, NLP

Used large language models and heterogeneous graph neural networks to predict drug-target interactions

- 4th at ISEF, 93.7% accuracy. Used LLM's to generate embeddings for drugs and proteins via SMILES and Amino acid sequences. Built a heterogeneous graph neural network to represent relationships.

Awards/Honors

- 4th Place Grand Award at ISEF, in Computational Biology and Bioinformatics - Regeneron, 2023
- 1st Place in Computational Biology - Washington State Science and Engineering Fair 2023
- Certificate of Achievement - Wolfram Computation Meets Knowledge, 2023

Skills

Languages: Python(Strong), Java(Strong), C(Intermediate), C++(Intermediate), R, Javascript

Tools/Misc: PyTorch, SLURM, ROS Noetic, LLMs, VLAs, Numpy, English/Mandarin, Spinning pens.